



Introduction to Pathology

Lec. 1

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Definition of pathology

Pathology : scientific study of changes in the structure and function of the body in diseases .

- Pathology as a word means:
- Patho-logy : the study (*logos*) of suffering (*pathos*).



Branches of pathology

General Pathology

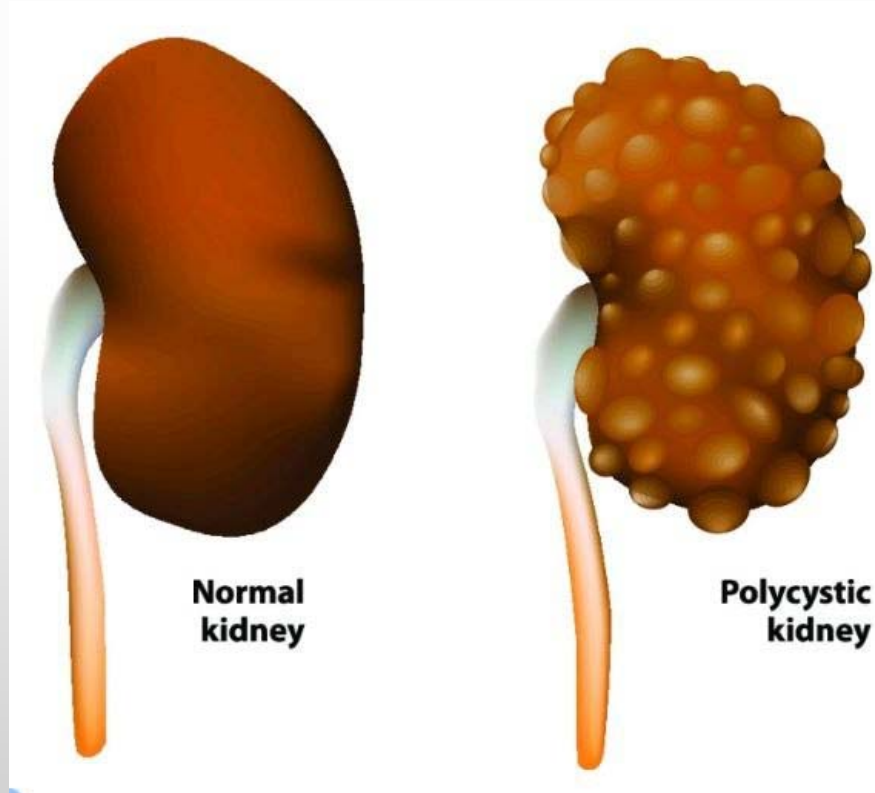
Study of the common changes in all tissues. e.g. inflammation, cancer, edema, hemorrhageetc.

Systemic Pathology

Discussing the pathologic mechanisms in relation to various organ systems e.g. cardiovascular system, respiratory system , urinary system.....etc.

What is the Disease?

- Abnormal alteration of structure or function in any part of the body.



Genetically determined diseases: abnormality in the DNA of the fertilized ovum that is inherited from one or both parents (thalassemia).

Acquired diseases: occurs due to environmental factors like:
Inflammatory – Trauma, infections, immune

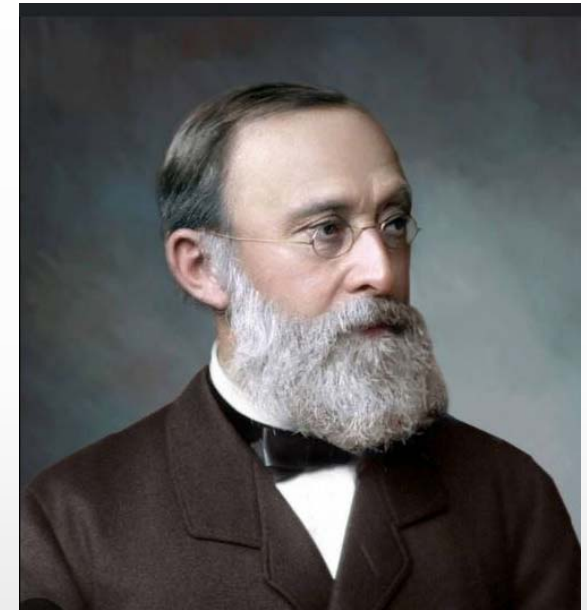
- **Pathology involves the investigation of:**

- 1- **Etiology**: refers to the underlying causes and modifying factors that are responsible for the initiation and progression of disease .
- 2- **Pathogenesis**: The underlying mechanisms that result in the presenting signs and symptoms of the disease. Thus, etiology refers to **why** a disease arises and pathogenesis describes **how** a disease develops.
- 3- **Morphology**: Identify changes in the gross or microscopic appearance of cells, tissues and organs.

- Defining the etiology and pathogenesis of disease not only is essential for understanding disease but also is the basis for developing rational treatments and effective preventive measures. Thus, pathology provides the scientific foundation for the practice of medicine.

- **Rudolph Virchow 1821-1902, the Father of Modern Pathology**

- The primitive light microscope of Rudolf Virchow (1821-1902), a German pathologist, enabled him to see changes in diseased tissues at a cellular level. His observations have had a profound influence on the understanding of many diseases.



Methods used to study pathology

- 1- **Histopathology**: Examination of the diseased tissues by light or electron microscope.
- 2- **Cytopathology**: Examination of isolated cells for diagnosis of the disease.
- 3- **Biochemistry**: Examination of the metabolic disturbances of diseases by evaluation of various compounds in body fluids.
- 4- **Microbiology**: Identification of the causative microorganism of the disease by examination of body fluids, mucosal surfaces and excised tissues using microscopical, cultural and serological techniques.

5- **Haematology**: Microscopical examination of blood.

6- **Cytogenetics**: Examination of chromosomal abnormality.

7- **Histochemistry**: Detection of cells and tissues constituents using special immunohistochemical techniques.

8- **Forensic pathology**: Examination of tissues taken from autopsy (dead body).

9- **Toxicology**; concerned with the study of the effects of known or suspected poisons on the body.

Fields of pathology

- **Experimental pathology** refers to the observation of the effects of manipulations on animal models or cell cultures regarding researches on human diseases.
- **Clinical pathology:** the approach to the patient's illness clinically is based on the following sequence of steps :

Patient's history _Examination_Investigations _Diagnosis _Treatment

Important terms in pathology

- **Sign:** is an objective indication of some medical fact or quality that is detected by a physician during a physical examination of a patient , example,(pale of the face, skin rash).
- **Symptom:** is subjective indication, observed by the patient, example (fever, pain, headache).
- **Biopsy :** It is a piece of living tissue taken for histopathological examination. Many different types of tissue can be biopsied, including skin, bone, organs and other soft tissues.

- **Autopsy:** also known as a post-mortem examination or necropsy it is a procedure that consists of a thorough examination of a dead body to determine the cause and manner of death and to evaluate any disease or injury that may be present. It is usually performed by a specialized medical doctor called a pathologist.
- **Prognosis :** Is prediction of the disease course and outcome (prediction of the future of a disease) .
- Types of prognosis :
- **Good prognosis :** high chance of recovery as mild inflammations .
- **Poor prognosis :** low chance of recovery as in malignant tumors .